

MBB — ΔPPM development vs team context (exploratory spec)

Status: Charles request 2026-10-08 — document before code.
Audience: Charles, model chapter, COMPASS / implementation.
Related: `MBB_FIRST_SEASON_AT_SCHOOL_SORT_DECISIONS.md` (first@school PPM definition); Scholar VECTOR $A^{\text{final}} = B^{\text{entry}} + \Delta(\text{environment})$ framing; `3-Master_Plan/BINDING_Selection_is_its_own_step.md`.

One-sentence estimand

Among players with observable **first and last seasons at the same school**, describe how **college PPM growth** $\Delta\text{PPM} = \text{PPM}_{\text{last}} - \text{PPM}_{\text{first@school}}$ associates with **exit-year team / peer context** ($\hat{T}_j$, LOO pool quality, viable-peer congestion) and optionally `Y_draft` — **associational**, not causal congestion or hero replacement.

Motivation

- MBB has **no** high-school / recruiting B_i in the downloaded stack.
 - We **do** have **first season at school** and **last player-season (last-ps)** production from ESPN box → panel.
 - Reigning hero plots `P(Y=1)` vs `poolq_loo` at a **single time slice**; they do **not** measure **within-player development**.
 - This exploratory track asks: does **visible PPM growth** over a college spell **co-vary** with playing on **stronger teams** ($\hat{T}_j$) or **more congested viable-peer ponds** (`L_C` at last-ps)? Minutes enter as **opportunity** controls or companion plots.
- This is **not** assignment-time assortativity ($H_{\text{sort}}^{\text{entry}}$). It is **post-assignment** ΔA vs **exit-year** context.

Core quantities

Development (player, within school)

Symbol / column	Definition
<code>season_first@school</code>	<code>min(season)</code> over rows with same <code>(athlete_id, team_id)</code> after panel hygiene (dash names, <code>min_team_season_games</code> — same rule as sort memo).
<code>ppm_first</code>	Raw PPM on the first@school row: <code>points/minutes</code> (or panel <code>ppm</code>).
<code>season_last</code>	<code>max(season)</code> for that <code>(athlete_id, team_id)</code> (for one-and-done, first = last; exclude or flag).
<code>ppm_last</code>	Raw PPM on the last@school row.
<code>delta_ppm</code>	<code>ppm_last - ppm_first</code> (primary y-axis).
<code>minutes_first</code> , <code>minutes_last</code>	Minutes on first and last rows; optional <code>delta_minutes</code> .

Sensitivity (later): within-season z-PPM each year, then subtract; label `delta_ppm_z`.

Peer / team context (attach to last@school row unless noted)

Computed with **reigning same-season** `perf` (season PPM → optional within-season z) and existing LOO machinery — **not** freshman PPM for teammates.

X-axis option	Column / construction	Interpretation
Team mean $\hat{T}_j$	Mean <code>perf</code> on <code>(team_id, season_last)</code> roster (document include-self vs exclude-self).	"How strong was the team in the exit season?"
LOO quality	<code>poolq_loo</code> on last@school row	Mean teammate <code>perf</code> (congestion-adjacent quality , not viable count).
Viable-peer congestion	<code>congestion_crowding</code> or <code>pool_c_smooth_loo</code> via <code>tier1_mechanism_vars</code> / Tier 1 θ , γ	"How packed with above- θ peers?" (L_C).

Outcome (descriptive overlay)

Field	Use
<code>Y_draft</code>	Color, facet, or separate panels — outcome , not regressor for "development mechanism" claims.

Analysis grain

Primary (recommended): one row per `(athlete_id, team_id)` where `season_first@school < season_last` (at least two seasons at that program).

Reigning-aligned export for draft plots: merge Δ columns onto the last-ps cross-section (one row per athlete at global `max(season)`); if the player transferred, last-ps team may differ from the school where Δ was computed — document two modes:

Mode	Grain	When to use
A — within-school	<code>(athlete_id, team_id)</code> , require 2+ seasons	Cleanest “development at this program.”
B — last-ps athlete	One row per athlete at last college season; Δ from last team only or longest spell	Align with draft-year hero population (define rule explicitly).

Default for v1 implementation: Mode A; Mode B only if Charles selects for draft-facing figures.

Filters (proposed defaults — align with hero where possible)

Filter	Default	Notes
Season window	2009–2021 (reigning)	Configurable.
<code>min_team_season_games</code>	10	On first and last season rows (or both seasons in spell).
Minutes floor	20 on both first@school and last@school	Symmetric gate; drops noisy PPM both ends.
One-and-done	Exclude <code>season_first == season_last</code>	Or separate “zero Δ ” bin.
<code>perf</code> for LOO	PPM → within-season z (<code>perf_zscore_within_season=True</code>) on full panel before LOO	Match Pass A / BDP last-ps prep unless sensitivity says raw.

Report N after each filter (survival table in provenance).

Figures (v1)

Not replacements for reigning HERO PNGs. Write under a disposable sandbox, e.g. `sports_sandbox/_DISPOSABLE_delta_ppm_development/` or `sports/exports/delta_ppm_dev/`.

- Scatter or hexbin: `delta_ppm` (y) vs $\hat{T}_j$ (x), points colored by `Y_draft` (alpha for overlap).
- Same y, x = `poolq_loo` (last season).
- Same y, x = `congestion_crowding` or `pool_c_smooth_loo` (last season, θ from drafted-perf rule used elsewhere).
- Companion: `delta_minutes` vs same x-axes, or partial view — `delta_ppm` vs x within ventiles of `minutes_first`.
- Optional binned means: ventiles of x → mean `delta_ppm` ± CI (readable alternative to raw scatter).

Caption boilerplate: Associational; exit-year peer measures; Δ PPM confounded by role, minutes, selection into last-ps; not HS talent; not causal effect of congestion on draft.

What this is NOT

- Not B_i or recruiting rank.
- Not $H_{\text{sort}}^{\text{entry}}$.
- Not the hero estimand (`P(Y=1)` binned on `poolq_loo` only).
- Not score `S_i = A_i - \lambda L_C` — unless a later explicit extension puts Δ into an outcome equation with new auth.

Implementation sketch (when authorized)

- `prepare_panel` → full player-season panel (box rebuild or 530 CSV).
- `apply_perf_metric_for_analysis` on all rows (for LOO on last season).
- Build `first_last_at_school` table: join first and last rows on `(athlete_id, team_id)` → `delta_ppm`, minutes.
- Attach last-row `poolq_loo`, $\hat{T}_j$, `add_tier1_mechanism_variables` congestion columns.
- Apply filters; write `delta_ppm_development_panel.csv` + PNGs + `provenance.json` (filters, N, θ , season window).

Reuse: `panel_build`, `y_draft_mode.restrict_to_last_season_rows` (for Mode B), `tier1_mechanism_vars`, BDP-style $\hat{T}_j$ from `bdp_ai_tj_distributions` if already standardized.

Do not overwrite `player_season_panel_530.csv` or reigning `pass_a/` artifacts.

Open decisions (Charles)

#	Question	Default if silent
O1	Raw vs z for <code>delta_ppm</code>	Raw primary; z sensitivity
O2	$\hat{T}_j$ include self?	Match existing BDP convention (document in provenance)
O3	Mode A vs B for draft-colored plots	Mode A v1
O4	Congestion x-axis	<code>pool_c_smooth_loo</code> + hard share sensitivity
O5	θ for viable peers	Same as 539 / <code>viability_theta_drafted_perf</code> median rule

Thread log

Date	Entry
2026-10-08	Initial spec from Charles: Δ PPM first@school $\rightarrow$ last@school vs $\hat{T}_j$, LOO, congestion; minutes as opportunity; scatter with <code>Y_draft</code> ; exploratory only.